

# Curriculum Vitae: Zongnan Li

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## PERSONAL INFORMATION

Nationality: China

Institute: Korean Astronomy and Space Science Institute, South Korea

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Language: Chinese (native), English (proficient), Japanese (N4)

## EDUCATION/EMPLOYMENT

**Korean Astronomy and Space Science Institute (KASI)** Daejeon, South Korea

*East Asian Core Observatories Association (EACOA) postdoc, Astrophysics*

Host: Dr. Jongsoo Kim

January 2026 – present

**National Astronomical Observatory of Japan (NAOJ)** Tokyo, Japan

*East Asian Core Observatories Association (EACOA) postdoc, Astrophysics*

Host: Dr. Daisuke Iono

April 2024 – Dec. 2025

**National Astronomical Observatories, CAS (NAOC)** Beijing, China

*Postdoc, Fellowship of China National Postdoctoral Program for Innovative Talents, Astrophysics*

Host: Dr. Di Li

Aug 2022 – March 2024

**Nanjing University**

Nanjing, China

*Ph.D., Astrophysics*

Sep 2017 – June 2022

PhD thesis: Galactic Circumnuclear Environments in M31 and M81: A Multi-wavelength, High-definition Study

Advisor: Dr. Zhiyuan Li

**Nanjing University**

Nanjing, China

*B.S., Astronomy*

Sep 2013 – June 2017

## RESEARCH INTERESTS

**Galactic nuclei, SMBH-galaxy co-evolution, Interstellar medium, Star formation**

## OBSERVING PROPOSALS

- 2025, **PI**, **MeerKAT**, 2.5 hours.

“Mapping the polarized synchrotron emission from the central region of M31”

- 2025, **PI**, **ALMA**, 23.5 hours.

“Probing Molecular Gas in Massive Group-central Disk Galaxies: Toward Understanding Star Formation Quenching”

- 2025, **PI**, **ALMA**, 22.3 hours.

“Revealing the molecular gas environment in and surrounding the nuclear star cluster in M33”

- 2025, **PI**, **JCMT**, 26 hours.

“JCMT CO(3-2) mapping of a peculiar ionized gas feature in the M31 disk”

- 2025, **PI**, **Nobeyama**, 50 hours.

“TIANGANG-Nobeyama: probing molecular gas content in nearby Milky Way analogs”

- 2025, Co-**PI**, **WHT/WEAVE**, 4 hours.

“How Does Stellar Feedback WEAVE into the ISM: the Unique Case of M31”

- 2025, Co-I, **EP/FXT**, 75 ks.

*“Exploring the Hot Gaseous Halo of NGC 3521”*

- 2025, Co-I, **CFHT/SITELLE**, 6 hours.

*“Illuminating the Outburst: A SITELLE View of M31’s Turbulent Past”*

- 2025, Co-I, **VLA**, 4.65 hours.

*“Unveiling the Radio Counterpart of Galactic-Scale Bubbles in NGC 6286”*

- 2024, Co-**PI**, **WHT/WEAVE**, 4 hours.

*“Tracing the Circumnuclear Ionized Gas in M81 with WEAVE”*

- 2023, **PI**, **P200/CWI**, 1 night.

*“Spectroscopic Mapping of A Triple Galaxy System Caught in the Act of Merging”*

- 2023, **PI**, **CFHT/SITELLE**, 4 hours.

*“Smoking Gun of A kpc-scale Outflow in the Disk of M31?”*

- 2023, **PI**, **VLA**, 19.6 hours.

*“Probing the cold neutral hydrogen in the M31/M33 halo”*

- 2023, **PI**, **VLA**, 11 hours.

*“Exploring the Imprint of An Energetic Wind from A Prototype Low-Luminosity AGN”*

- 2023, Co-**PI**, **NOEMA**, 8 hours.

*“Probing Molecular Gas in and around the Nuclear Star Cluster of M33”*

- 2023, **PI**, **JCMT/M24AP024**, 8 hours in band 1 and 21 hours in band 3.

*“Extending JCMT CO(3-2) mapping of the nuclear ring of M31”*

- 2023, Co-**PI**, **WHT/WEAVE**, 4 hours.

*“A Spectroscopic View of Young Star Clusters and H II regions in M31”*

- 2022, **PI**, **PCWI/CTAP2023-A0038**, 1 night.

*“Spectroscopic Mapping of A Triple Galaxy System Caught in the Act of Merging”*

- 2022, Co-I, **IRAM NOEMA/W22CQ**, 4 + 8 hours at Grade B.

*“Probing Circumnuclear Gas Inflows/Outflows in Merging Galaxy Pairs”*

- 2022, Co-I, **GTC/GTC02-22ADDT**, 2 hours.

*“Probing the ionized nuclear outflow in M81”*

- 2020, **PI**, **JCMT/M19BP006**, 19 hours in band 3 and 40 hours in band 4 at Tier 1.

*“JCMT CO(3-2) mapping of the nuclear ring of M31”*

- 2020, **PI**, **IRAM-30m/053-20**, 51 hours at Grade B.

*“Mapping molecular gas in the third closest massive spiral galaxy M81”*

- 2019, **PI**, **IRAM-30m/185-19 & 058-19**, 15 + 22 hours at Grade B.

*“IRAM 30m CO(1-0) mapping of the nuclear ring of M31”*

- 2018, Co-I, **IRAM-30m/194-18**, 17 hours at Grade B.

*“Probing Molecular Gas in Massive Central Disk Galaxies: Toward Understanding Star Formation Quenching”*

- 2017, Co-I, **CAHA-3.5m**, 2 hours.

*“CAHA/PPAK IFU mapping of M81”*

- 2017, Co-I, **JCMT/M17BP005**, 276 hours, Large program.

*“HARP and SCUBA-2 High-Resolution Terahertz Andromeda Galaxy Survey”*

## ACADEMIC VISITS

- **University of California, Los Angeles (UCLA)**

Host: Dr. Mark Morris

*Ionization mechanisms of nearest LINERs M31 and M81*

- **Cardiff University**

Host: Dr. Stephen Eales

Los Angeles, USA

Aug 2025 – Aug 2025

Cardiff, UK

Jun 2023 – Jun 2023

*Molecular cloud properties across M31*

- **Instituto de Astrofísica de Andalucía (CSIC)**

Granada, Spain

Host: Dr. Ruben Garcia-Benito

Jan 2019 – Jan 2019

*CAHA/PPAK Integral-field Spectroscopic Observations of M81*

- **IRAM 30m telescope**

Sierra Nevada, Spain

Jan 2019 – Jan 2019

Conduct observation for program: Probing Molecular Gas in Massive Central Disk Galaxies

- **James Clerk Maxwell Telescope (JCMT)**

Hilo, Hawaii, USA

Oct 2017 – Oct 2017

Conduct observation for HASHTAG program

- **Harvard and Smithsonian Center for Astrophysics**  
USA

Cambridge, MA,

Host: Dr. Qizhou Zhang

Feb 2017 – May 2017

*Structure and Kinematics of the young planetary nebula NGC 7027*

- **University of Wisconsin-Madison**

Madison, WI, USA

Host: Dr. Eric Hooper

Jun 2016 - Aug 2016

*Discovering and Age-dating AGN Radio Emission in Large-scale Galaxy Surveys*

ORAL PRESENTATIONS

- “Understanding circumnuclear environments in M31 and M81 with spatially-resolved multiphase gas”

**The 11th East-Asian Meeting on Astronomy**, Niigata, Japan, Dec., 2025

- “Understanding Galactic Circumnuclear Environments in M31 and M81 with spatially resolved multiphase gas”

**Multiphase ISM seminar**, Nagoya, Japan, August, 2025

- “Are molecular clouds stable in M31?”

**WE-Heraeus-Seminar, Charting the Cosmos**, Germany, August, 2025

- (Invited talk) “Ring or no ring - Revisiting the Multiphase Nuclear Environment in M31”

**On-line lunch talk**, Yunnan University, China, Mar. 2024

- “Revisiting the Nuclear Environment of our neighbor M31- A tale of molecular gas”

**East Asian Young Astronomers Meeting**, Chiang Mai, Thailand, Feb. 2024

- “Circumnuclear gas in M81: a unique testbed for AGN and stellar photoionization”

**Resolving galaxies in all scales**, Hong Kong, China, Dec. 2023

- “Revisiting the Nuclear Ring in our Neighborhood: --a JCMT perspective of M31”

**JCMT Users meeting**, London, UK, May 2023

- (Invited talk) “Understanding Galactic Circumnuclear Environments in M31 and M81 with Multi-wavelength, High-definition Observations”

**NAOJ colloquium**, Tokyo, Japan, Apr. 2023

- (Invited talk) “Cold gas in the nuclear region (~500 pc) of M31”

**JCMT Users meeting**, EAO, on-line talk, Feb. 2022

- “Ionized gas in the nuclear region of M81 -- CAHA IFU mapping of the central 500 pc”

**Seminar on diffuse gas in nearby galaxies**, Xiamen University, China, May 2021

- (Invited talk) “Cold gas in the nuclear region (~500 pc) of the Andromeda galaxy -- CO and [CII] mapping of selected regions in M31”

**On-line lunch talk**, Yunnan University, China, Mar. 2021

- “JCMT CO(3-2) mapping of M31”

**JCMT/SMA User's Meeting**, ASIAA, Taipei, Dec 2019

- “JCMT CO(3-2) Mapping of the disk and Circumnuclear Region of M31”  
**EAO future meeting**, Purple Mountain Observatory, China, May 2019
- “JCMT mapping of CO(3-2) in the circumnuclear region of M31”  
**Academic annual meeting of Chinese Astronomical Society**, Yunnan University, China, Oct 2018

#### AWARDS AND HONORS

- **East Asian Core Observatories Association (EACOA) fellowship** EACOA, 2023
- **KIAA Postdoctoral Fellowship (declined)** KIAA, 2022
- **China National Postdoctoral Program for Innovative Talents**  
China Postdoctoral Science Foundation, 2022

#### COMPUTER SKILL

- Python, LaTeX, Shell, C, Fortran
- Various astronomy software (e.g., CASA, Starlink, GILDAS, starlight, DS9, IRAF, pPXF, CLOUDY, HIPE)

#### PRESS RELEASE

- JCMT newsletter – “Molecular gas at the center of Andromeda, M31”  
<https://www.eaobservatory.org/jcmt/wp-content/uploads/sites/2/2019/09/EAO-NEWS-5.pdf>

#### PROFESSIONAL ACTIVITIES

- SOC/LOC member of “The Andromeda Galaxy in Multiple wavelength seminar”, Suzhou, China, 2024
- SOC member of “East Asian Young Astronomers Meeting”, Chiang Mai, Thailand, 2024
- LOC member of “NAOC-CUHK ISM science forum”, Beijing, China, 2022

#### PUBLICATION LIST

- Lead-author publications: 6, Corresponding author: 1, Co-author: 8
1. Ruling out Conventional Photoionization Models in the Closest LINER, M31, with CFHT/SITELLE Observations  
**Li, Z.-N.**, Li, Z., Wang, S., et al. 2026, ApJ, 999, 213
  2. Ring or No Ring—Revisiting the Multiphase Nuclear Environment in M31  
**Li, Z.-N.**, Su, Z., Wang, S., et al. 2025, ApJ, 938, 128
  3. CAHA/PPAK Integral-field Spectroscopic Observations of M81. II. Testing Photoionization Models in a Spatially Resolved LINER  
**Li, Z.-N.**, Li, Z., Garcia-Benito, R., 2023, ApJ, 958, 89
  4. CAHA/PPAK Integral-field Spectroscopic Observations of M81 – I. Circumnuclear Ionized Gas

- Li, Z.-N.**, Li, Z., Garcia-Benito, R., et al. 2022, ApJ, 928, 111
5. A Herschel Mapping of [CII], [OI], and [OIII] Lines from the Circumnuclear Region of M31  
**Li, Z.-N.**, Li, Z., Smith, M. W. L., and Gao, Y., 2020, ApJ, 905, 135
  6. The HASHTAG project I. A survey of CO(3-2) emission from the star forming disc of M31  
**Li, Z.-N.**, Li, Z., Smith, M. W. L., et al. 2020, MNRAS, 492, 195
  7. JCMT mapping of CO(3-2) in the circumnuclear region of M31  
**Li, Z.-N.**, Li, Z., Zhou, P., et al. 2019, MNRAS, 484, 964
  8. The HASHTAG project II. Giant molecular cloud properties across the M31 disc  
Deng, Y., **Li, Z.-N.**, Li, Z., et al. 2025, MNRAS, 538, 2445
  9. Deep Andromeda JCMT SCUBA-2 Observations. The Submillimeter Maps and Giant Molecular Clouds  
Jiao, S., Wu, J., Liu, H. B., Tsai, C., Lin, Y., ..., **Li, Z.-N.**, et al., ApJS, 2026, 282, 39
  10. Discovery of giant bubbles in the hot gaseous halo of the massive disk galaxy NGC 6286  
Lin, H., Li, Z., **Li, Z.-N.**, 2025, ApJ, 992, 86.
  11. Wind-fed Supermassive Black Hole Accretion by the Nuclear Star Cluster: the Case of M31\*  
Su, Z., Li, Z., **Li, Z.-N.**, 2025, ApJ, 988, 68S
  12. Episodic Feedback in Triple Active Galactic Nucleus Candidate SDSS J0849+1114 Revealed by Extended Ionized Gas  
Xu, X., Hou, M., Li, Z., Peng, S., Su, Z., **Li, Z.-N.**, et al. 2025, ApJ, 993, 35
  13. The ALMaQUEST Survey XV: The Dependence of the Molecular-to-Atomic Gas Ratios on Resolved Optical Diagnostics  
Yu, N., Zheng, Z., et al. **Li, Z.-N.**, et al. 2024, SCPMA, 67, 299811
  14. NOEMA Detection of Circumnuclear Molecular Gas in X-ray-selected Dual Active Galactic Nuclei: No Evidence for Heavy Obscuration  
Hou, M., Li, Z., Liu, X., **Li, Z.-N.**, Li, R., et al. 2023, ApJ, 943, 50
  15. Quenching of Massive Disk Galaxies in the IllustrisTNG Simulation  
Xu, Y., Luo, Y., Kang, X., Li, Z., **Li, Z.-N.**, Wang, P., Libeskind, N., 2022, ApJ, 928, 100
  16. The HASHTAG project: The First Submillimeter Images of the Andromeda Galaxy from the Ground  
Smith, M. W. L., Eales, S. A., Williams, T. G., Lee, B., **Li, Z.-N.**, et al. 2021, ApJS, 257, 52S
  17. What has quenched the massive spiral galaxies?

Luo, Y., **Li, Z.-N.**, Kang, X., Li, Z., Wang, P. 2020, MNRAS, 496, L116

18. The Physical Properties of S0 Galaxy PGC 26 218 : The Origin of Starburst and Star Formation

Ge X., Gu Q.-S., Garcia-Benito R., Xiao M.-Y., **Li Z.-N.**, 2020, ApJ, 889, 132

†: corresponding author.